

Inequalities, Bounds & Equality Conditions

Integrated Assimilation Book

For a learner who remembers some inequality formulas but does not yet consistently separate bound, equality, and attainment.
Central belief: A bound is NOT automatically the requested extremum.

1. RECONNECT - three different statements

For real x ,

$$(x-3)^2 \geq 0.$$

Therefore

$$x^2 - 6x + 13 \geq 4.$$

This proves 4 is a lower bound. Because equality occurs at $x=3$, and 3 is an allowed real number, the minimum is 4.

Now change only the domain to $x > 3$.

The same lower bound 4 is still true, but equality would require $x=3$, which is not allowed. So on $x > 3$, 4 is not a minimum.

That one contrast contains the whole topic:

BOUND $\rightarrow$ EQUALITY $\rightarrow$ ATTAINMENT.

First attempt

Decide whether each conclusion is justified.

1. $x^2 > 0$ for $x \neq 0$, so the minimum over $x \neq 0$ is 0.
2. For $x > 0$, AM-GM gives $x + 9/x \geq 6$; therefore the minimum is 6.
3. For integer n , $(n-1/2)^2 + 2 \geq 2$; therefore the integer minimum is 2.

Do not calculate first. Inspect the equality case and the domain.

2. DISCOVER - read the request and direction before choosing a tool

A question asking for a minimum needs a lower-bound route. A question asking for a maximum needs an upper-bound route.

This sounds obvious, but many incorrect solutions prove a true inequality in the wrong direction.

Contrast

For nonnegative x, y with $x+y=10$:

$$(x-y)^2 \geq 0$$

implies

$$(x+y)^2 \geq 4xy, \text{ so } xy \leq 25.$$

This is an upper bound on xy , useful for a maximum. It does not give a positive lower bound: xy can be 0 if zero is allowed, and can approach 0 without attaining it if $x, y > 0$.

Adopt the first four router questions

1. What is requested: minimum, maximum, feasibility, or just a proof?
2. What is the domain?
3. Is the expression bounded in the needed direction?
4. Which representation exposes that direction?

3. MAKE SENSE - completing the square is an optimization representation

A quadratic often reveals its bound immediately after completing the square.

Example: minimum

$$f(x) = x^2 - 6x + 13$$

becomes

$$f(x) = (x-3)^2 + 4.$$

Since a square is nonnegative, $f(x) \geq 4$, with equality at $x=3$.

Example: maximum

$$g(x) = 10 - (x-2)^2.$$

Since $(x-2)^2 \geq 0$,

$$g(x) \leq 10, \text{ with equality at } x=2.$$

Boundary with discriminant

If the request is "for which parameter values does a quadratic equation have real roots?", that is a root-feasibility question owned by the polynomial/root-feasibility topic. Do not turn every quadratic optimization problem into a discriminant problem.

4. DISCOVER - AM-GM is a bound plus an equality condition

For positive reals a, b ,

$$a+b \geq 2\sqrt{ab}.$$

Equality occurs exactly when $a=b$.

The hypotheses matter: positivity is not decoration.

Example

For $x>0$, find the minimum of

$$x+9/x.$$

Both terms are positive. AM-GM gives

$$x+9/x \geq 2\sqrt{9}=6.$$

Equality needs

$$x=9/x, \text{ so } x=3 \text{ (the positive solution).}$$

Because $x=3$ belongs to the domain, the minimum is 6.

Contrast: upper bound, not lower bound

If $a, b \geq 0$ and $a+b=12$, then

$$ab \leq 36, \text{ equality at } a=b=6.$$

So 36 is the maximum product. There is no reason to call it a minimum.

Contrast: positive but no minimum product

If $a, b > 0$ and $a+b=12$, then $ab > 0$, but values such as $a=0.1$, $b=11.9$ make the product small. It can approach 0, but never attain 0. Thus there is no positive minimum.

5. MAKE SENSE - Cauchy/Engel only when the structure justifies it

For positive x, y , the Engel form gives

$$1/x + 1/y \geq (1+1)^2/(x+y).$$

If $x+y=10$,

$$1/x+1/y \geq 4/10 = 2/5.$$

Equality in this case occurs at $x=y=5$, so the minimum is $2/5$.

The point is not to collect names of inequalities. The point is to recognize a representation in which the desired direction and equality condition are transparent.

Why not use Cauchy everywhere?

If square completion gives the target in one line, importing a stronger theorem adds cognitive cost without adding information.

6. DIAGNOSE - bound, equality, attainment, discrete filter

Error A - lower bound = minimum

For $x>0$, $x^2>0$, so 0 is a lower bound, but it is not attained. There is no minimum; the infimum is 0.

Error B - real equality point used in an integer problem

For real x ,

$$(x-1/2)^2+2 \geq 2, \text{ equality at } x=1/2.$$

For integer n , $1/2$ is inadmissible. Check the nearest integers:

$$n=0 \text{ or } 1 \text{ gives } 1/4+2=9/4.$$

So the integer minimum is $9/4$, not 2.

Error C - equality condition forgotten

An inequality theorem may prove the numerical bound correctly while the proposed equality case violates a given restriction. The correct conclusion then changes.

Error D - wrong direction

To maximize xy under fixed sum, proving $xy \geq 0$ is true but irrelevant.

7. TRY - support fades toward independence

Attempt before reading support.

Full support - execution supplied

For $x>0$, minimize $x+16/x$.

Use AM-GM:

$$x+16/x \geq 2\sqrt{16}=8, \text{ equality at } x=4.$$

Medium support - representation supplied

For real x , find the minimum of $x^2+8x+20$.

Cue: complete the square. Execute and verify equality yourself.

Light support - recognition clue

For positive a, b with $a+b=14$, maximize ab .

Clue: fixed sum, product target, equality should be balanced.

Independent - no route supplied

For integer n , find the minimum of n^2-5n+9 .

Choose the continuous representation, then apply the discrete filter.

8. ADOPT - the full optimization router

text

REQUEST

-> DOMAIN

-> BOUNDED?

-> DIRECTION

-> REPRESENTATION

-> BOUND

-> EQUALITY

-> ATTAINMENT

-> DISCRETE FILTER

-> CHECK

Stop conditions

- If the expression is unbounded in the requested direction, stop: no finite extremum.
- If equality is impossible in the domain, do not call the bound a minimum/maximum.
- If the domain is discrete, do not stop at a continuous equality point.

9. Historical anchor traces

IOQM-2025-Q07

The independently verified route writes the relevant sum as $n=x+y$ and uses

$$(x-y)^2 = 2(n+1012)-n^2 \geq 0.$$

This yields an upper feasibility bound on n ; the integer filter gives the largest admissible integer 46.

Pedagogical role: feasibility bound -> direction -> integer filter.

IOQM-2024-Q06

The power conditions force the variables into a very small discrete set; equality/extreme behavior can then be checked exactly. The independently verified answer is 06.

Pedagogical role: strong bound -> equality structure -> discrete admissibility.

These are validated anchors, not evidence of official topic frequency.

10. TRANSFER

Changed domain

A bound attained over reals may fail to be attained over positive reals, integers, or a punctured interval.

Geometry

A perimeter/area problem may reduce to a fixed-sum product bound. Geometry remains the historical owner when the geometric structure is decisive; this inequality topic supplies the optimization mechanism.

Number theory

A continuous inequality can create a short interval, after which integer/divisibility filters decide the actual answer.

Polynomial boundary

If the request is root count/real-root existence, retrieve reversible-transformation habits and route to the polynomial/root-feasibility topic. If the request is the quadratic's minimum value, stay with inequality optimization and complete the square.

Final belief

Prove the bound, name the equality condition, test attainment in the real domain of the question, then filter discrete candidates if necessary.

Inequalities, Bounds & Equality Conditions - First-Step Reference

One-question router

What exactly must be extreme, over which domain, and in which direction?

text

REQUEST -> DOMAIN -> BOUNDED? -> DIRECTION -> REPRESENTATION
-> BOUND -> EQUALITY -> ATTAINMENT -> DISCRETE FILTER -> CHECK

Recognition atlas

Visible structure | First useful question

quadratic expression | complete square?

positive terms with fixed product | AM-GM for lower bound?

fixed positive sum, product target | upper bound at balance?

reciprocal sum + fixed sum | Engel/Cauchy justified?

integer domain | where is real optimum, then which nearby integers?

strict domain / excluded equality point | is the bound attained?

root-existence request | route to the polynomial/root-feasibility method?

Equality checklist

1. What equality condition does the inequality require?
2. Does that condition satisfy every original constraint?
3. Is the equality point in the requested domain?
4. If not, is there another equality case?
5. If no equality is attainable, state bound/infimum/supremum correctly rather than inventing a minimum/maximum.

Mandatory contrasts

- lower bound vs minimum;
- upper bound vs maximum;
- real optimum vs integer optimum;
- inequality optimization vs discriminant/root feasibility.

Inequalities, Bounds & Equality Conditions - Recognition and First-Line Lab

Write only the first useful line or the first logical question.

1. Minimize $x^2 - 8x + 20$ over real x .
2. Maximize $12 - (x+1)^2$ over real x .
3. Minimize $x + 25/x$ for $x > 0$.
4. Maximize ab for $a, b \geq 0$, $a+b=18$.
5. Decide whether 0 is a minimum of x^2 on $x > 0$.
6. Minimize $(n-1/2)^2 + 3$ over integers n .
7. Minimize $1/x + 1/y$ for $x, y > 0$, $x+y=12$.
8. A proof gives $f(x) \geq 7$ with equality only at $x=0$, but the domain is $x > 0$. What must you conclude/check next?
9. A question asks how many real roots $x^2 + kx + 4 = 0$ has. Is this canonically an inequality-optimization problem?
10. A question asks for the minimum value of $x^2 + kx + 4$ as a function of x . Which representation is canonical here?
11. For positive a, b with $ab=9$, minimize $a+b$.
12. For positive a, b with $a+b=9$, is there a minimum positive value of ab ?
13. You have proved $xy \leq 16$ under the given constraints. What two checks remain before saying "maximum 16"?
14. The continuous optimum occurs at $x=3.7$, but x must be an integer. What is the next step?
15. $f(x) = 1 - 1/x$ on $x > 0$ has upper bound 1. Is 1 a maximum?
16. The request is "prove $(a+b)^2 \geq 4ab$." Do you need to claim an extremum at all?

Inequalities, Bounds & Equality Conditions - Practice and Transfer Bank

F0 - foundation

1. Write $x^2-6x+13$ in completed-square form and state its real minimum.
2. State the real maximum of $9-(x-4)^2$.
3. For $x>0$, use AM-GM to bound $x+4/x$ below.
4. For nonnegative a,b with $a+b=10$, bound ab above.
5. Explain why $x^2\geq 0$ does not by itself prove a minimum of 0 on $x>0$.

F1 - direct

6. Minimize $x+9/x$ for $x>0$.
7. Maximize ab for $a,b\geq 0$, $a+b=12$.
8. Minimize $1/x+1/y$ for $x,y>0$, $x+y=10$.
9. Find the real minimum of $x^2+10x+29$.
10. Find the integer minimum of $(n-1/2)^2+2$.

F2 - standard

11. For positive x,y with $xy=16$, minimize $x+y$.
12. For positive x,y with $x+y=20$, maximize xy .
13. Determine whether x^2 on the domain $x\neq 0$ has a minimum.
14. Determine whether $1-1/x$ on $x>0$ has a maximum.
15. For integer n , minimize n^2-5n+9 .

F3 - disguised

16. A rectangle has perimeter 40. Find its maximum possible area and state the equality condition.
17. Positive numbers p,q satisfy $pq=36$. Find the minimum of $2p+8q$ after choosing a useful representation/substitution.
18. If $x,y>0$ and $x+y=1$, minimize $1/x+1/y$.
19. A real t satisfies $t+1/t=5$, $t>0$. Give a lower bound on t^2+1/t^2 and then determine its exact value from the relation; distinguish the two statements.
20. For integers $m,n\geq 0$ with $m+n=15$, maximize mn .

F4 - preliminary-style

21. Real x,y satisfy $x^2+y^2=x+y+1012$. What is the largest possible integer value of $x+y$? (Mechanism analogue of the 2025 anchor; do not use the historical stem as an authored item.)
22. For positive a,b,c with $abc=1$, prove $a+b+c\geq 3$ and state all equality conditions.
23. For positive x,y with $x+y=8$, minimize $x^2/y + y^2/x$ by a justified inequality route.
24. For integer n , find the minimum of $n^2-7n+15$ and compare it with the real minimum.
25. Determine whether the set $\{x+1/x : x>1\}$ has a minimum; justify attainment rather than quoting a bound alone.

Transfer

26. Representation change: solve the same real minimum $x^2-6x+13$ by square completion, then explain why a discriminant route would answer a different natural question.
27. Domain change: change the domain of x^2 from real x to $x>0$ to nonzero integers. State how the extremum conclusion changes each time.
28. Number-theory surface: an inequality reduces an integer variable to $10\leq n\leq 14$. Explain what inequality has accomplished and what remains for the integer owner.
29. Geometry surface: fixed perimeter gives a fixed sum of side lengths. Explain why the area maximum is the inequality/attainment mechanism inside a geometry surface.
30. Mixed transfer: create an example where a true lower bound is not a minimum because equality is excluded.

Inequalities, Bounds & Equality Conditions - Independent Mixed Mastery Check

No method labels or default hints.

1. Find the real minimum of $x^2-12x+40$.
2. Find the real maximum of $17-(x-5)^2$.
3. For $x>0$, find the minimum of $x+36/x$.
4. Positive a,b satisfy $a+b=16$. Find the maximum of ab .
5. Positive x,y satisfy $x+y=14$. Find the minimum of $1/x+1/y$.
6. Decide whether 0 is a minimum of x^2 on $x \neq 0$.
7. Find the integer minimum of $(n-3/2)^2+1$.
8. For integer n , find the minimum of $n^2-9n+23$.
9. A proof gives $F \geq 10$; equality requires $x=y$, but the original condition says $x \neq y$. State the strongest justified extremum conclusion without additional information.
10. For nonnegative p,q , $p+q=18$. Find the maximum of pq and verify attainment.
11. For positive p,q , $p+q=18$. Does pq have a minimum? Justify.
12. The request is "for which k does $x^2+kx+9=0$ have a repeated real root?" Route it to the correct canonical topic and state why this inequality topic should not duplicate the method.
13. A continuous relaxation has minimum 5 at $x=2.4$, but x must be integral. Describe the exact next calculation required before claiming the integer minimum.
14. A rectangle with perimeter 28 has sides $x,y>0$. Find the maximum area.
15. WHY NOT: a student writes AM-GM $a+b \geq 2\sqrt{ab}$ to maximize $a+b$ when only $ab=9$ is fixed. Diagnose why this does not establish a maximum.
16. Transfer: construct a function/domain pair where a finite supremum exists but no maximum exists.